

# Report

Data Analytics | Orange Data Mining | Tableau Visualisation

Subject: Data Visualisation

## 1. Dataset

### 1.1 Dataset Source Link

Definition: A dataset source link is the URL from which the raw data file was obtained. It ensures traceability, reproducibility, and allows evaluators to verify the authenticity of the data used in the project.

Dataset Link: [Future of Work in the Age of AI \(2020–2026\)](#)

### 1.2 Screenshot of Dataset Download

This screenshot confirms that the dataset was successfully accessed and downloaded from the source. It serves as proof of the original data acquisition step.

|    | A      | B          | C         | D         | E    | F            | G             | H     | I         | J          | K         | L      | M          | N     | O      | P         | Q         | R     | S      | T        | U          |                         |
|----|--------|------------|-----------|-----------|------|--------------|---------------|-------|-----------|------------|-----------|--------|------------|-------|--------|-----------|-----------|-------|--------|----------|------------|-------------------------|
| 1  | job_id | job_role   | industry  | country   | year | automatic_ai | replace_skill | gap   | salary_be | salary_aft | salary_ch | skill  | dem.remote | fr_ai | adopti | education | automatic | skill | trans  | wage_vol | reskilling | ai_disruption_intensity |
| 2  | 0      | Data Anal  | Technolog | Canada    | 2021 | 26.22        | 30.94         | 73.2  | 101839    | 99454.4    | -2.34     | 2.66   | 15.23      | 86.62 | 2      | Low       | 49.71     | 2.34  | 33.15  | 22.7118  |            |                         |
| 3  | 1      | Accountai  | Finance   | Brazil    | 2020 | 52.08        | 56.41         | 2.06  | 146389    | 139517     | -4.69     | 10.43  | 26.36      | 18.34 | 5      | Medium    | 27.07     | 4.69  | 22.857 | 9.55147  |            |                         |
| 4  | 2      | Teacher    | Technolog | USA       | 2020 | 31.3         | 31.61         | 43.19 | 64947.5   | 58369.4    | -10.13    | 8.14   | 36.29      | 36.64 | 2      | Medium    | 37.245    | 10.13 | 28.516 | 11.4683  |            |                         |
| 5  | 3      | Customer   | Technolog | Brazil    | 2021 | 56.92        | 63.42         | 19.97 | 91708.1   | 86715.7    | -5.44     | 6.11   | 64.68      | 17.05 | 5      | Medium    | 38.445    | 5.44  | 30.391 | 9.70486  |            |                         |
| 6  | 4      | Teacher    | Manufact  | Japan     | 2024 | 14.55        | 17.17         | 96.56 | 127008    | 119379     | -6.01     | 2.08   | 71.58      | 44.02 | 3      | Low       | 55.555    | 6.01  | 36.591 | 6.40491  |            |                         |
| 7  | 5      | Software   | Healthcar | Singapore | 2022 | 8.54         | 8.53          | 3.44  | 139118    | 150562     | 8.23      | -7.21  | 38.05      | 52.01 | 3      | Low       | 5.99      | 8.23  | 6.917  | 4.44165  |            |                         |
| 8  | 6      | Accountai  | Manufact  | Australia | 2020 | 48.27        | 42.18         | 96.96 | 123016    | 132100     | 7.38      | 6.71   | 17.96      | 19.6  | 1      | Medium    | 72.615    | 7.38  | 50.61  | 9.46092  |            |                         |
| 9  | 7      | Teacher    | Finance   | Australia | 2026 | 13.17        | 12.25         | 38.87 | 62561.9   | 59680      | -4.61     | 15.57  | 35.28      | 54.27 | 3      | Low       | 26.02     | 4.61  | 18.312 | 7.14736  |            |                         |
| 10 | 8      | Marketing  | Retail    | USA       | 2025 | 19.86        | 22.27         | 7.46  | 148426    | 138379     | -6.77     | 11.12  | 10.5       | 81.55 | 4      | Low       | 13.66     | 6.77  | 12.213 | 16.1958  |            |                         |
| 11 | 9      | Financial  | Finance   | Singapore | 2020 | 59.48        | 64.93         | 77.13 | 38885.4   | 35622      | -8.39     | 1.91   | 87.68      | 62.33 | 5      | Medium    | 68.305    | 8.39  | 49.448 | 37.0739  |            |                         |
| 12 | 10     | HR Manag   | Education | Germany   | 2025 | 33.16        | 27.37         | 31.1  | 69022     | 74630.2    | 8.13      | 18.56  | 89.85      | 47.22 | 1      | Medium    | 32.13     | 8.13  | 25.033 | 15.6582  |            |                         |
| 13 | 11     | Software   | Manufact  | Canada    | 2020 | 8.37         | 9.09          | 76.08 | 97353.3   | 97004.5    | -0.36     | 20.65  | 57.05      | 42.75 | 2      | Low       | 42.225    | 0.36  | 26.28  | 3.57817  |            |                         |
| 14 | 12     | Data Anal  | Transport | Canada    | 2023 | 5            | 4.22          | 3.14  | 106369    | 107345     | 0.92      | -14.88 | 91.68      | 24.93 | 3      | Low       | 4.07      | 0.92  | 3.218  | 1.2465   |            |                         |
| 15 | 13     | Accountai  | Education | Australia | 2021 | 38.73        | 42.69         | 22.88 | 39237.6   | 36065.3    | -8.08     | -0.02  | 93.67      | 80.81 | 3      | Medium    | 30.805    | 8.08  | 24.78  | 31.2977  |            |                         |
| 16 | 14     | Data Anal  | Healthcar | Brazil    | 2025 | 44.34        | 50.93         | 80.37 | 52388.4   | 52897      | 0.97      | 14.69  | 82.67      | 89.61 | 2      | Medium    | 62.355    | 0.97  | 42.138 | 39.7331  |            |                         |
| 17 | 15     | Accountai  | Energy    | Singapore | 2022 | 32.26        | 27.23         | 22.79 | 81252.9   | 83659      | 2.96      | 7.61   | 10.63      | 51.07 | 5      | Medium    | 27.525    | 2.96  | 20.629 | 16.4752  |            |                         |
| 18 | 16     | Customer   | Education | USA       | 2021 | 54.22        | 48.19         | 11.99 | 70513.8   | 69376.5    | -1.61     | 9.04   | 56.69      | 70.3  | 1      | Medium    | 33.105    | 1.61  | 25.768 | 38.1167  |            |                         |
| 19 | 17     | Mechanic   | Transport | Canada    | 2020 | 35.45        | 42.14         | 96.24 | 60213.9   | 48660.2    | -19.19    | 4.73   | 35.64      | 3.69  | 2      | Medium    | 65.845    | 19.19 | 48.809 | 1.30811  |            |                         |
| 20 | 18     | Marketing  | Education | Germany   | 2025 | 52.67        | 52.73         | 5.15  | 63437.6   | 62217.3    | -1.92     | 8.02   | 23.04      | 48.95 | 4      | Medium    | 28.91     | 1.92  | 23.189 | 25.782   |            |                         |
| 21 | 19     | Teacher    | Energy    | UK        | 2022 | 79           | 70.84         | 67.21 | 121394    | 130997     | 7.91      | -4.09  | 43.1       | 63.23 | 2      | High      | 73.105    | 7.91  | 54.136 | 49.9517  |            |                         |
| 22 | 20     | Customer   | Retail    | Singapore | 2024 | 69.35        | 70.34         | 9.03  | 130236    | 117336     | -9.91     | -0.66  | 13.67      | 59.09 | 4      | High      | 39.19     | 9.91  | 33.422 | 40.9789  |            |                         |
| 23 | 21     | Mechanic   | Manufact  | UK        | 2024 | 57.43        | 46.32         | 51.21 | 57179.5   | 51105.3    | -10.62    | 9.74   | 72.18      | 38.67 | 4      | Medium    | 54.32     | 10.62 | 41.521 | 22.2082  |            |                         |
| 24 | 22     | Data Anal  | Technolog | India     | 2021 | 65.57        | 56.06         | 34.11 | 43616.8   | 47165.1    | 8.14      | -7.31  | 83.55      | 55.52 | 2      | High      | 49.84     | 8.14  | 38.903 | 36.4045  |            |                         |
| 25 | 23     | Teacher    | Finance   | Singapore | 2023 | 47.08        | 42.21         | 9.31  | 137666    | 141244     | 2.6       | 12.82  | 40.51      | 34.92 | 5      | Medium    | 28.195    | 2.6   | 22.405 | 16.4403  |            |                         |
| 26 | 24     | Truck Driv | Retail    | Brazil    | 2026 | 75.82        | 87.86         | 88.71 | 123585    | 115181     | -6.8      | 7.32   | 24.55      | 89.86 | 1      | High      | 82.265    | 6.8   | 58.981 | 68.1319  |            |                         |
| 27 | 25     | Data Anal  | Retail    | Australia | 2020 | 47.45        | 34.12         | 10.15 | 109620    | 130073     | 18.66     | 9.74   | 68.68      | 72.43 | 3      | Medium    | 26.3      | 18.66 | 25.673 | 9.52154  |            |                         |

## **2. Orange Data Mining**

### **2.1 What is Orange?**

Orange is an open-source data mining and machine learning tool built on Python. It provides a visual programming environment where users can drag and drop widgets to build data analysis workflows without writing code. Each widget performs a specific function such as loading data, cleaning it, visualising it, or applying machine learning algorithms. Orange is widely used for educational and exploratory data analysis purposes because of its interactive and intuitive interface.

### **2.2 Operations Performed**

The following operations were carried out in Orange on the Future of work (AI Job Replacement) dataset:

- Data Preprocessing
- Data Continuize
- Data Discretize
- Data Randomize
- Data Workflow

### **2.3 Screenshots of Orange Operations (Minimum 3–5 Required)**

#### **Screenshot 1 – Preprocess**

**A tool used to clean and prepare data by applying different changes before using it in models.**

The image displays two screenshots of a data visualization tool, likely Tableau, showing a dataset of job roles and salaries. The top screenshot shows the 'Info' tab, and the bottom screenshot shows the 'Variables' tab. Both screenshots feature a red box highlighting the 'Info' section.

**Top Screenshot (Info Tab):**

- Info:** 15000 instances (no missing data), 20 features, No target variable, No meta attributes.
- Variables:** Show variable labels (if present), Visualize numeric values, Color by instance classes.
- Selection:** Select full rows.

**Bottom Screenshot (Variables Tab):**

- Variables:** Show variable labels (if present), Visualize numeric values, Color by instance classes.
- Selection:** Select full rows.

**Dataset Columns:**

| job_id | job_role          | industry       | country   | year | promotion_risk_percent | replacement_cost | skill_gap_index | salary_before_usd | salary_after_usd |
|--------|-------------------|----------------|-----------|------|------------------------|------------------|-----------------|-------------------|------------------|
| 0      | Data Analyst      | Technology     | Canada    | 2021 | 26.22                  | 30.94            | 73.20           | 101819.02         | 99454            |
| 1      | Accountant        | Finance        | Brazil    | 2020 | 52.08                  | 56.41            | 2.06            | 146389.18         | 139516           |
| 2      | Teacher           | Technology     | USA       | 2020 | 31.30                  | 31.61            | 43.19           | 64647.50          | 56369            |
| 3      | Customer Supp...  | Technology     | Brazil    | 2021 | 56.92                  | 63.42            | 19.97           | 91706.13          | 86715            |
| 4      | Teacher           | Manufacturing  | Japan     | 2024 | 14.55                  | 17.17            | 96.56           | 127007.68         | 119379           |
| 5      | Software Engin... | Healthcare     | Singapore | 2022 | 8.54                   | 8.53             | 3.44            | 139118.45         | 150561           |
| 6      | Accountant        | Manufacturing  | Australia | 2020 | 48.27                  | 42.18            | 96.96           | 123015.94         | 132100           |
| 7      | Teacher           | Finance        | Australia | 2026 | 13.17                  | 12.25            | 38.87           | 62561.88          | 59680            |
| 8      | Marketing Spec... | Retail         | USA       | 2025 | 19.86                  | 22.27            | 7.46            | 148426.43         | 138379           |
| 9      | Financial Analyst | Finance        | Singapore | 2020 | 59.48                  | 64.93            | 77.13           | 38885.36          | 35622            |
| 10     | HR Manager        | Education      | Germany   | 2025 | 33.16                  | 27.37            | 31.10           | 69022.00          | 74630            |
| 11     | Software Engin... | Manufacturing  | Canada    | 2020 | 8.37                   | 9.09             | 76.08           | 97353.26          | 97004            |
| 12     | Data Analyst      | Transportation | Canada    | 2023 | 5.00                   | 4.22             | 3.14            | 106369.25         | 107345           |
| 13     | Accountant        | Education      | Australia | 2021 | 38.73                  | 42.69            | 22.88           | 39237.59          | 36065            |
| 14     | Data Analyst      | Healthcare     | Brazil    | 2025 | 44.34                  | 50.93            | 80.37           | 52388.41          | 52896            |
| 15     | Accountant        | Energy         | Singapore | 2022 | 32.26                  | 27.23            | 22.79           | 87252.93          | 83659            |
| 16     | Customer Supp...  | Education      | USA       | 2021 | 54.22                  | 48.19            | 11.99           | 70513.82          | 69176            |
| 17     | Mechanical Eng... | Transportation | Canada    | 2020 | 35.45                  | 42.14            | 96.24           | 60213.88          | 48660            |
| 18     | Marketing Spec... | Education      | Germany   | 2025 | 52.67                  | 52.73            | 5.15            | 63437.58          | 62217            |
| 19     | Teacher           | Energy         | UK        | 2022 | 79.00                  | 70.84            | 67.21           | 121394.35         | 130997           |
| 20     | Customer Supp...  | Retail         | Singapore | 2024 | 69.35                  | 70.34            | 9.03            | 130236.30         | 117335           |
| 21     | Mechanical Eng... | Manufacturing  | UK        | 2024 | 57.43                  | 46.32            | 51.21           | 57179.49          | 51105            |
| 22     | Data Analyst      | Technology     | India     | 2021 | 65.57                  | 56.06            | 34.11           | 43616.82          | 47165            |
| 23     | Teacher           | Finance        | Singapore | 2023 | 47.08                  | 42.21            | 9.31            | 137665.89         | 141241           |
| 24     | Truck Driver      | Retail         | Brazil    | 2026 | 75.82                  | 87.86            | 88.71           | 127585.07         | 115180           |
| 25     | Data Analyst      | Retail         | Australia | 2020 | 42.45                  | 34.12            | 10.15           | 109620.21         | 130072           |
| 26     | Teacher           | Healthcare     | Japan     | 2020 | 59.85                  | 53.56            | 32.54           | 119578.97         | 131098           |

## Screenshot 2 – Continue

Converts text (categorical data) into numbers so machine learning models can understand it.



The screenshot displays the Orange3 data mining interface. On the left, the 'Discretize' widget is configured with 'Natural binning, desired bins: 3'. The 'ai\_adoption\_level' variable is selected for discretization. The right side shows two 'Data Table' widgets. The top widget displays the original data with columns: 'note\_feasibility\_sc', 'ai\_adoption\_level', 'education\_requirement', and 'education\_risk\_cat'. The bottom widget shows the same data after discretization, with 'ai\_adoption\_level' values replaced by ranges like '≥ 80', '20 - 40', and '40 - 60'.

**Discretize - Orange**

★ Default setting: keep

Filter...

- job\_id (default)
- year (default)
- automation\_risk\_percent (default)
- ai\_replacement\_score (default)
- skill\_gap\_index (default)
- salary\_before\_usd (default)
- salary\_after\_usd (default)
- salary\_change\_percent (default)
- remote\_feasibility\_score (default)
- ai\_adoption\_level (binning, desired=3): 20.00, ...**
- education\_requirement\_level (default)
- skill\_transition\_pressure (default)
- wage\_volatility\_index (default)
- reskilling\_urgency\_score (default)
- ai\_disruption\_intensity (default)

Keep numeric

Remove

Natural binning, desired bins: 3

Fixed width:

Time interval: year(s)

Equal frequency, intervals: 3

Equal width, intervals: 5

Entropy vs. MDL

Custom: e.g. 0.0, 0.5, 1.0 CC

Use default setting

Apply Automatically

**Data Table - Orange**

Info

15000 instances (no missing data)

20 features

No target variable.

No meta attributes.

Variables

Show variable labels (if present)

Visualize numeric values

Color by instance classes

Selection

Clear Selection

Restore Original Order

Send Automatically

|    | note_feasibility_sc | ai_adoption_level | education_requirement | education_risk_cat |
|----|---------------------|-------------------|-----------------------|--------------------|
| 1  | 15.23               | 86.62             | 2                     | Low                |
| 2  | 26.36               | 18.34             | 5                     | Medium             |
| 3  | 36.29               | 36.64             | 2                     | Medium             |
| 4  | 64.68               | 17.05             | 5                     | Medium             |
| 5  | 71.58               | 44.02             | 3                     | Low                |
| 6  | 38.05               | 52.01             | 3                     | Low                |
| 7  | 17.96               | 19.60             | 1                     | Medium             |
| 8  | 35.28               | 54.27             | 3                     | Low                |
| 9  | 10.50               | 81.55             | 4                     | Low                |
| 10 | 87.68               | 62.33             | 5                     | Medium             |
| 11 | 89.85               | 47.22             | 1                     | Medium             |

**Data Table (1) - Orange**

Info

15000 instances (no missing data)

20 features

No target variable.

No meta attributes.

Variables

Show variable labels (if present)

Visualize numeric values

Color by instance classes

Selection

Clear Selection

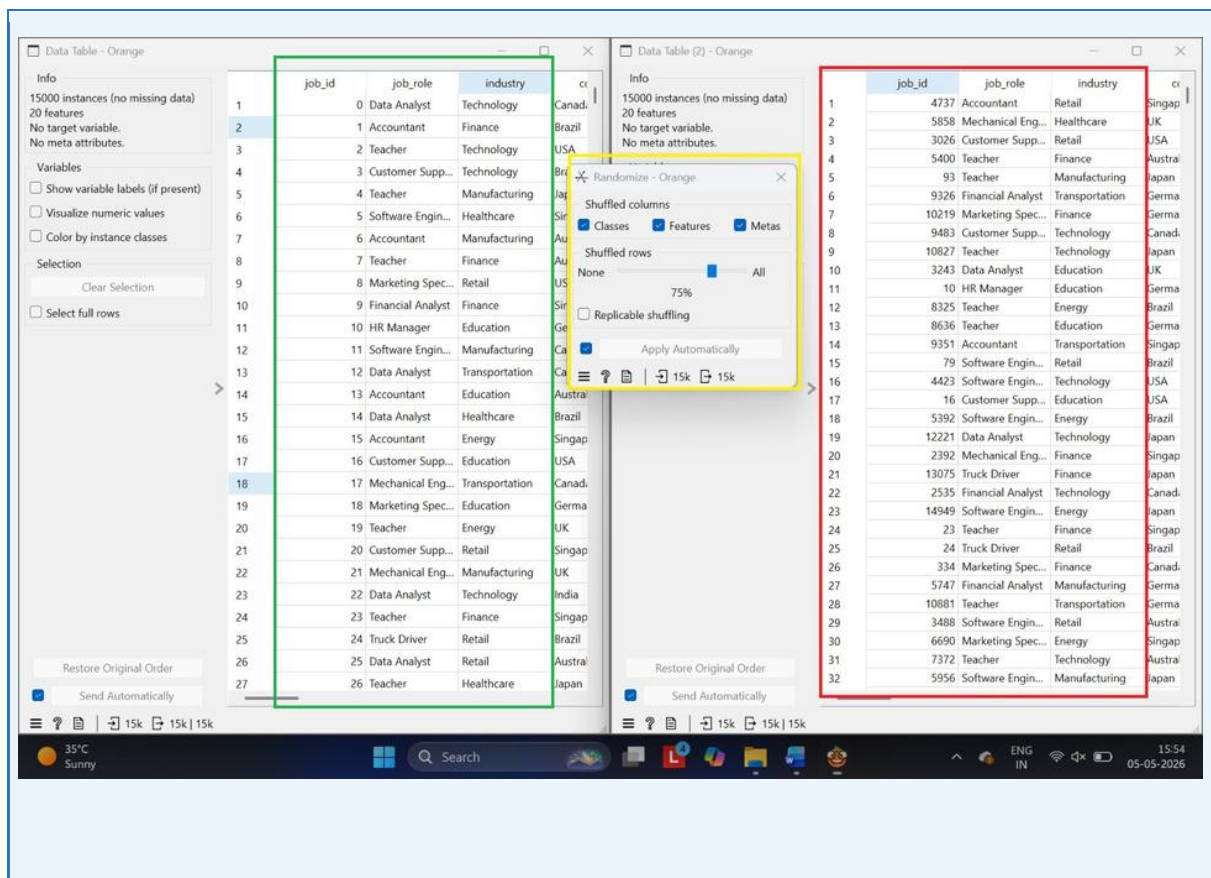
Restore Original Order

Send Automatically

|    | note_feasibility_sc | ai_adoption_level | education_requirement | education_risk_cat |
|----|---------------------|-------------------|-----------------------|--------------------|
| 1  | 15.23               | ≥ 80              | 2                     | Low                |
| 2  | 26.36               | < 20              | 5                     | Medium             |
| 3  | 36.29               | 20 - 40           | 2                     | Medium             |
| 4  | 64.68               | < 20              | 5                     | Medium             |
| 5  | 71.58               | 40 - 60           | 3                     | Low                |
| 6  | 38.05               | 40 - 60           | 3                     | Low                |
| 7  | 17.96               | < 20              | 1                     | Medium             |
| 8  | 35.28               | 40 - 60           | 3                     | Low                |
| 9  | 10.50               | ≥ 80              | 4                     | Low                |
| 10 | 87.68               | 60 - 80           | 5                     | Medium             |
| 11 | 89.85               | 40 - 60           | 1                     | Medium             |

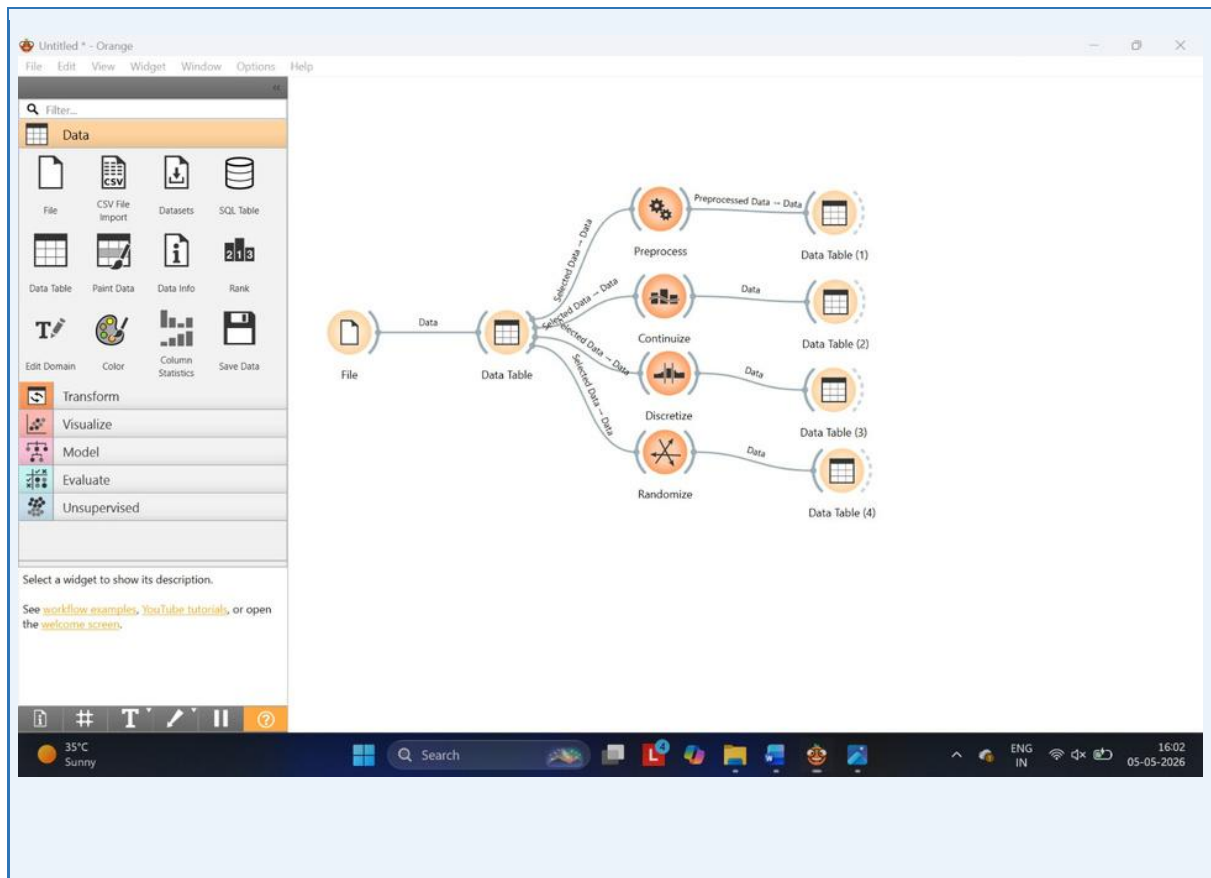
## Screenshot 4 – Randomize

Shuffles the data rows so the model doesn't get biased by the order of data.



## Screenshot 5 – Workflow

Orange is used to analyze data, build machine learning models, and visualize results without heavy coding.



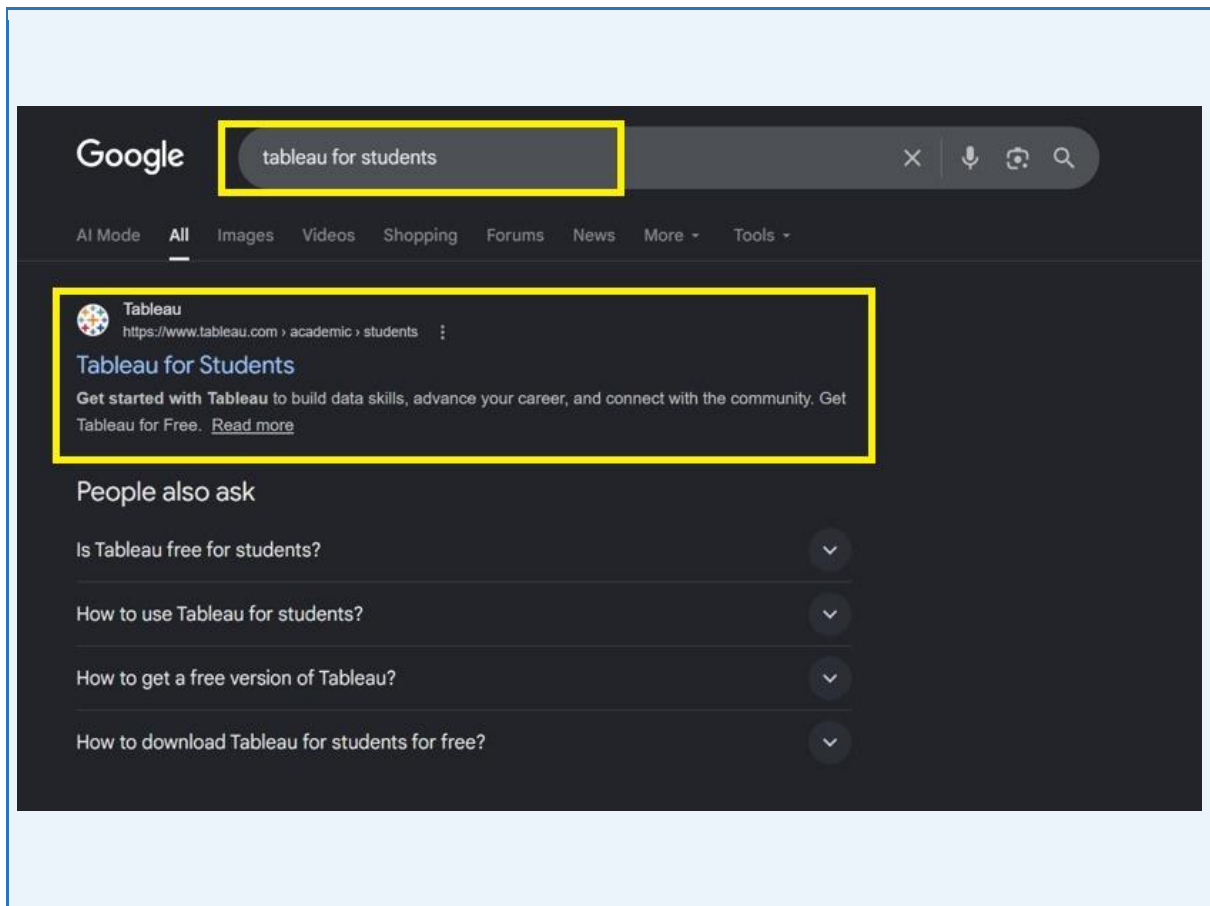
## 3. Tableau

### 3.1 Tableau Installation

Definition: Tableau is a powerful data visualisation platform that enables users to connect to various data sources and create interactive, shareable dashboards. It translates raw data into visual formats such as charts, graphs, and maps, making it easier for non-technical stakeholders to understand complex data patterns.

*Search " Tableau for Students " on Google .*

*Click on very first link as shown above .*



After the new webpage opens .

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## 3.2 Importing Dataset into Tableau

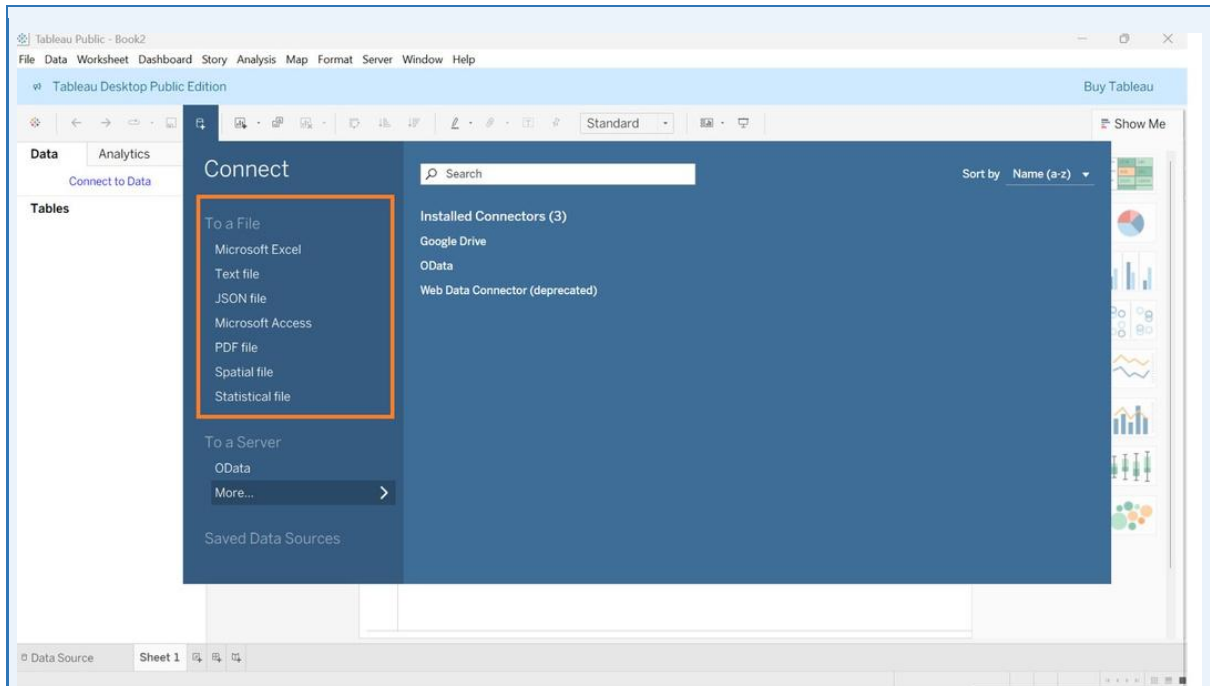
Definition: Importing a dataset in Tableau refers to connecting Tableau to a data source (such as an Excel file, CSV, or database) so that the data can be used for analysis and visualisation. This is done through the "Connect" pane on the Tableau Start Page. Once connected, Tableau loads the data into its Data Source editor where the user can preview and configure the data before building views.

Select appropriate type to open such as :

Excel to open excel type file

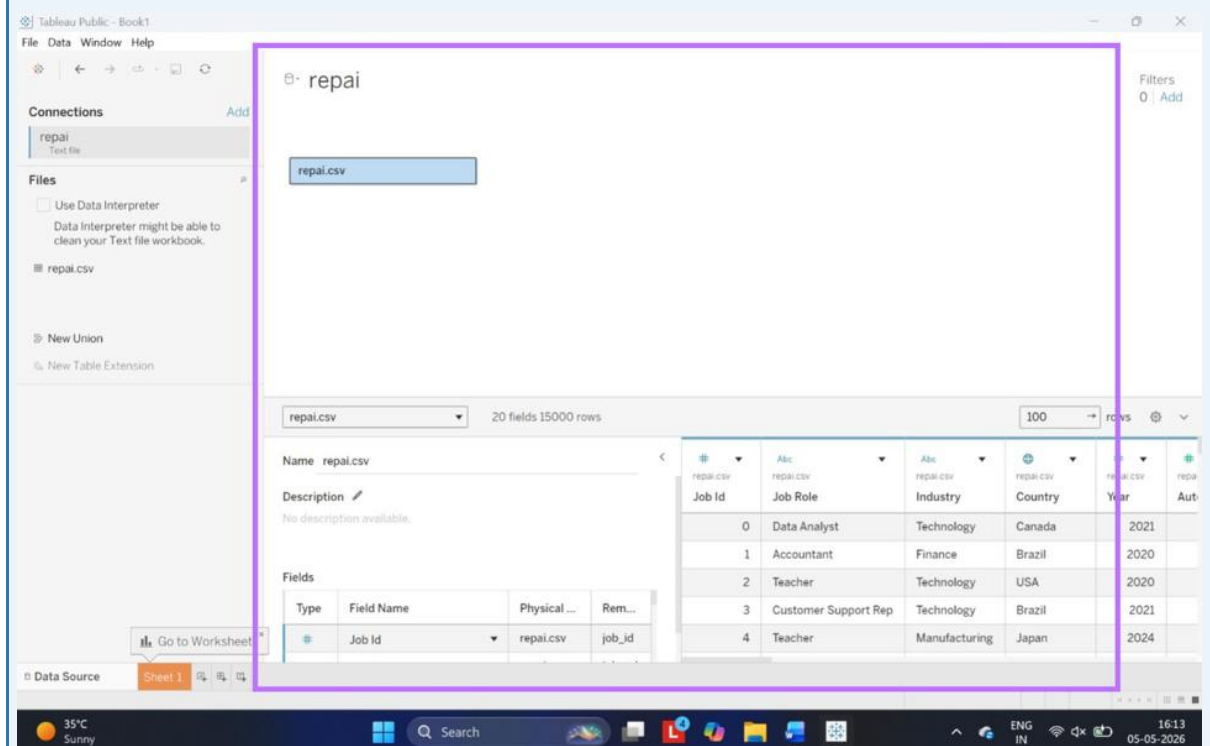
Text to open CSV type file

JSON to open key value pair type file .



*As above , your selected file type would open .*

*You can now perform various operation in tableau now .*





### 3.3 Joins in Tableau

Definition: A Join in Tableau is a method of combining data from two or more tables based on a related column (key field). Tableau supports four standard SQL join types, each determining which rows are included in the result set. Joins are performed in the Data Source editor by dragging a second table onto the canvas and selecting the join type.

#### Inner Join

Definition: An Inner Join returns only the rows where there is a matching value in both the left and right tables. Rows without a match in either table are excluded. It is the most commonly used join type and is useful when you only want records that exist in both datasets.

*Formula logic: Result = Left Table  $\cap$  Right Table*

The screenshot shows the Tableau Data Source editor. At the top, a message states "Sheet1 is made of 2 tables." Below this, two tables from "Sheet1" are connected by an inner join icon (two overlapping circles). A red box highlights the join configuration panel, which shows the join type set to "Inner". The primary table has a field named "Country" selected as the key, and the secondary table also has "country ..." selected. Below the join configuration, the main view displays a table with 25 fields and 25022606 rows. The table has columns for "Salary After Usd", "Salary Change Percent", "job id (Sheet1)", "job role (Sheet1)", and "industry (Sheet1)". The first few rows of data are visible, showing values for these fields.

#### Left Join

Definition: A Left Join returns all rows from the left (primary) table and the matched rows from the right (secondary) table. If there is no match in the right table, the result will contain NULL values for those columns. It is useful when you want to retain all records from the main dataset regardless of whether a matching record exists in the second table.

*Formula logic: Result = All of Left Table + Matching rows from Right Table*

Sheet1 is made of 2 tables. ⓘ

Sheet1

Join

Inner Left Right Full Outer

Data Source Sheet11

Country = country ...

Add new join...

Sheet1 25 fields 25022606 rows 100 rows

Name Sheet1

Description No description available.

Fields

| Type            | Field Name | Phys...  | Rem...   |
|-----------------|------------|----------|----------|
| #               | Job Id     | Sheet1   | job_id   |
| Abc             | Job Role   | Sheet1   | job_role |
| Go to Worksheet | Sheet1     | industry |          |

| #   | Sheet1            | Sheet1           | Sheet1                | Sheet11          | Abc Sheet11        |
|-----|-------------------|------------------|-----------------------|------------------|--------------------|
|     | Salary Before Usd | Salary After Usd | Salary Change Percent | job id (Sheet11) | job role (Sheet11) |
| 500 | 74,458.30         | 76,568.31        | 2.8300                | 0                | Data Analyst       |
| 700 | 38,682.39         | 36,961.90        | -4.4500               | 0                | Data Analyst       |
| 500 | 71,205.97         | 75,614.00        | 6.1900                | 0                | Data Analyst       |
| 500 | 127,446.60        | 121,745.92       | -4.4700               | 0                | Data Analyst       |
| 300 | 32,985.75         | 33,782.33        | 2.4100                | 0                | Data Analyst       |
| 200 | 37,096.96         | 37,121.45        | 0.0700                | 0                | Data Analyst       |
| 900 | 96,527.83         | 112,078.21       | 16.1100               | 0                | Data Analyst       |

## Right Join

Definition: A Right Join returns all rows from the right (secondary) table and the matched rows from the left (primary) table. If there is no match in the left table, NULL values appear for those columns. It is the mirror image of a Left Join and is useful when the secondary dataset must be fully preserved.

*Formula logic: Result = Matching rows from Left Table + All of Right Table*

Sheet1 is made of 2 tables. ⓘ

Sheet1

Join

Inner Left Right Full Outer

Data Source Sheet11

Country = country ...

Add new join...

Sheet1 25 fields 25022606 rows 100 rows

Name Sheet1

Description No description available.

Fields

| Type            | Field Name | Phys...  | Rem...   |
|-----------------|------------|----------|----------|
| #               | Job Id     | Sheet1   | job_id   |
| Abc             | Job Role   | Sheet1   | job_role |
| Go to Worksheet | Sheet1     | industry |          |

| #   | Sheet1            | Sheet1           | Sheet1                | Sheet11          | Abc Sheet11        |
|-----|-------------------|------------------|-----------------------|------------------|--------------------|
|     | Salary Before Usd | Salary After Usd | Salary Change Percent | job id (Sheet11) | job role (Sheet11) |
| 500 | 74,458.30         | 76,568.31        | 2.8300                | 0                | Data Analyst       |
| 700 | 38,682.39         | 36,961.90        | -4.4500               | 0                | Data Analyst       |
| 500 | 71,205.97         | 75,614.00        | 6.1900                | 0                | Data Analyst       |
| 500 | 127,446.60        | 121,745.92       | -4.4700               | 0                | Data Analyst       |
| 300 | 32,985.75         | 33,782.33        | 2.4100                | 0                | Data Analyst       |
| 200 | 37,096.96         | 37,121.45        | 0.0700                | 0                | Data Analyst       |
| 900 | 96,527.83         | 112,078.21       | 16.1100               | 0                | Data Analyst       |

## Full Outer Join

Definition: A Full Outer Join returns all rows from both the left and right tables. Where there is no match, NULL values are inserted for the missing columns. It is useful when you want a complete picture of both datasets, including unmatched records from either side.

*Formula logic: Result = All of Left Table  $\cup$  All of Right Table*

The screenshot shows a data tool interface. At the top, a window titled "Sheet1 is made of 2 tables." displays a join configuration. Two "Sheet1" tables are selected, and the "Full Outer" join type is chosen. The join key is set to "Country". Below this, a table with 25 fields and 25022606 rows is shown. The table has columns for "Salary After Usd", "Salary Change Percent", "job id (Sheet1)", "job role (Sheet1)", and "industry (Sheet1)". The data is displayed in a grid with 10 rows visible.

| # | Sheet1           | Sheet1                | Sheet1          | Sheet1            | Sheet1            |
|---|------------------|-----------------------|-----------------|-------------------|-------------------|
|   | Salary After Usd | Salary Change Percent | job id (Sheet1) | job role (Sheet1) | industry (Sheet1) |
| 0 | 76,568.31        | 2.8300                | 0               | Data Analyst      | Technology        |
| 9 | 36,961.90        | -4.4500               | 0               | Data Analyst      | Technology        |
| 7 | 75,614.00        | 6.1900                | 0               | Data Analyst      | Technology        |
| 0 | 121,745.92       | -4.4700               | 0               | Data Analyst      | Technology        |
| 5 | 33,782.33        | 2.4100                | 0               | Data Analyst      | Technology        |
| 6 | 37,121.45        | 0.0700                | 0               | Data Analyst      | Technology        |
| 3 | 112,078.21       | 16.1100               | 0               | Data Analyst      | Technology        |

### 3.4 Grains & Charts (5 Grains + 5 Charts)

**Definition – Grain:** The grain (or granularity) of a dataset refers to the level of detail represented by a single row in the data. For example, if each row represents one employee, the grain is "one record per employee." Understanding grain is critical before building charts, as it determines what can be meaningfully aggregated or compared.

#### Grain 1 – Job Role per Country

**Grain Description:** The grain represents the intersection of a specific professional role and a specific geographic location. It captures how many individuals (represented by the count) within a country belong to a specific job category, such as "Software Engineer" or "Data Scientist."

#### Chart 1 – Stacked Bar Chart

**Definition:** A Stacked Bar Chart is a visualization that uses bars to show the total value of a category, while dividing that bar into colored segments to represent the contribution of different sub-groups. In this case, the total height is the sum of all countries for that role, and each colored block is an individual country.

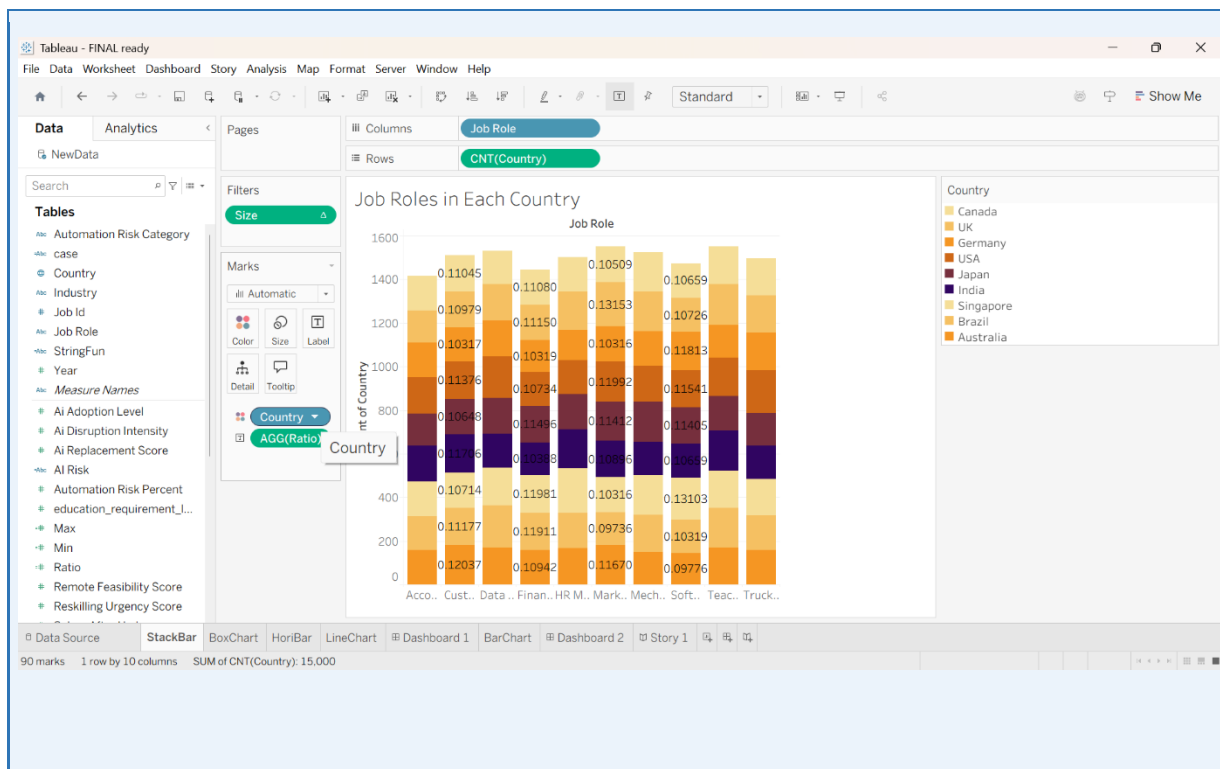
**Purpose in Project:** The purpose of this chart is to compare the total headcount across various industries while simultaneously visualizing the geographic composition of each. It allows users to see which job roles are most prevalent overall (e.g., "Teacher"). It helps identify if certain roles have a higher concentration of people from specific countries (indicated by the decimal labels showing the ratio/percentage).

**Specifications:**

X-Axis: Job Role (Categorical labels like Accountant, Data Scientist, etc.)

Y-Axis: Count of Country (CNT)

(A quantitative measure representing the total frequency/count of records, ranging from 0 to 1600).



## Grain 2 – Industry by Salary Change Percent

**Grain Description:** The grain of this data is the individual distribution of salary changes within specific industries. Each industry (e.g., Education, Healthcare, Tech) contains multiple data points representing different salary change percentages, which are then summarized to show the spread, median, and outliers.

## Chart 2 – Box Plot

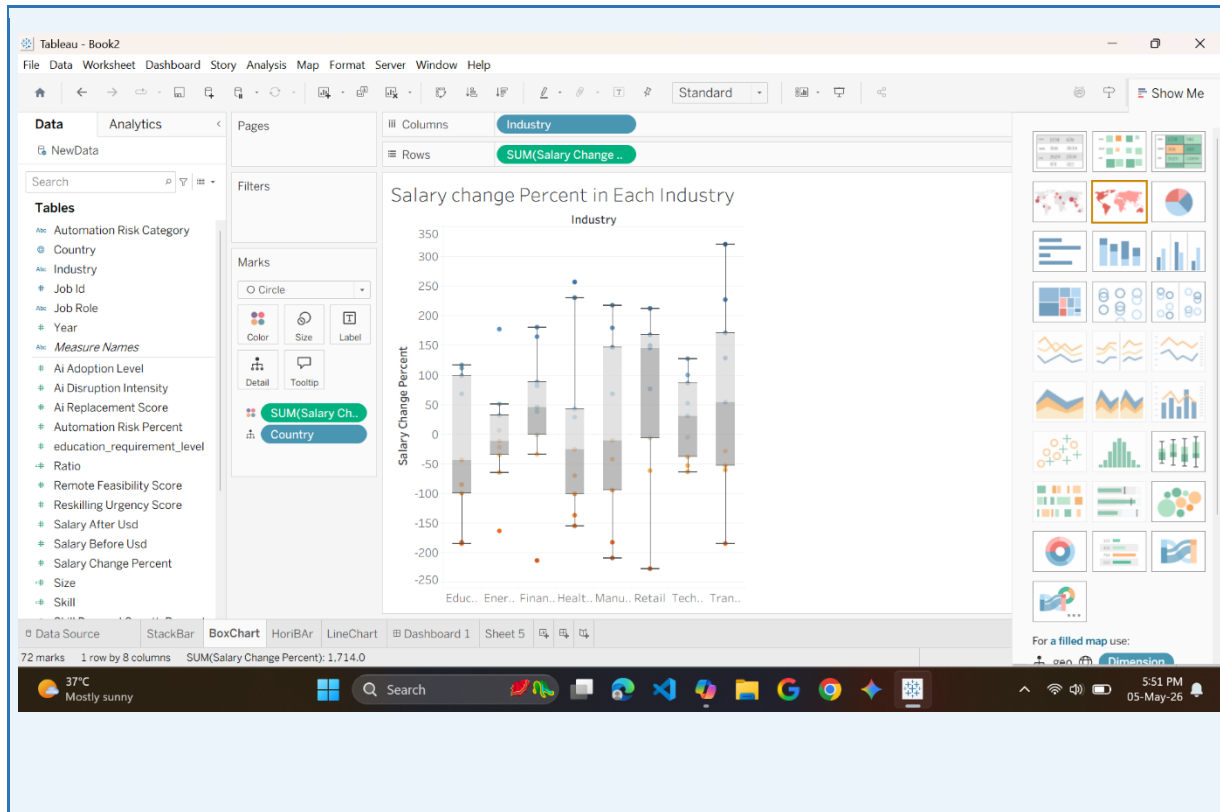
**Definition:** A Box and Whisker Plot is a standardized way of displaying the distribution of data based on a five-number summary: the minimum, first quartile (Q1), median, third quartile (Q3), and maximum. It highlights the "interquartile range" (the box) and identifies outliers (the dots outside the whiskers).

**Purpose in Project:** The purpose of this chart is to analyze the volatility and range of salary changes across different sectors. It identifies which industries have a consistent salary change (shorter boxes) versus those with high variance (tall boxes like "Retail" or "Transportation"). It specifically helps in spotting outliers—unusual cases where salary changes were extremely high (above 300%) or extremely low (below -200%).

**Specifications:**

X-Axis: Industry (Categorical / Discrete groups such as Education, Energy, Finance, etc.)

Y-Axis: Salary Change Percent (Quantitative / Continuous measure ranging from -250 to 350)



### Grain 3 – Job Role by Skill Demand Growth

Grain Description: The grain of this data is a specific professional category (Job Role) mapped against its calculated demand growth. Each row represents a unique job title, showing its individual performance in terms of market or skill demands.

### Chart 3 – Horizontal Bar Chart

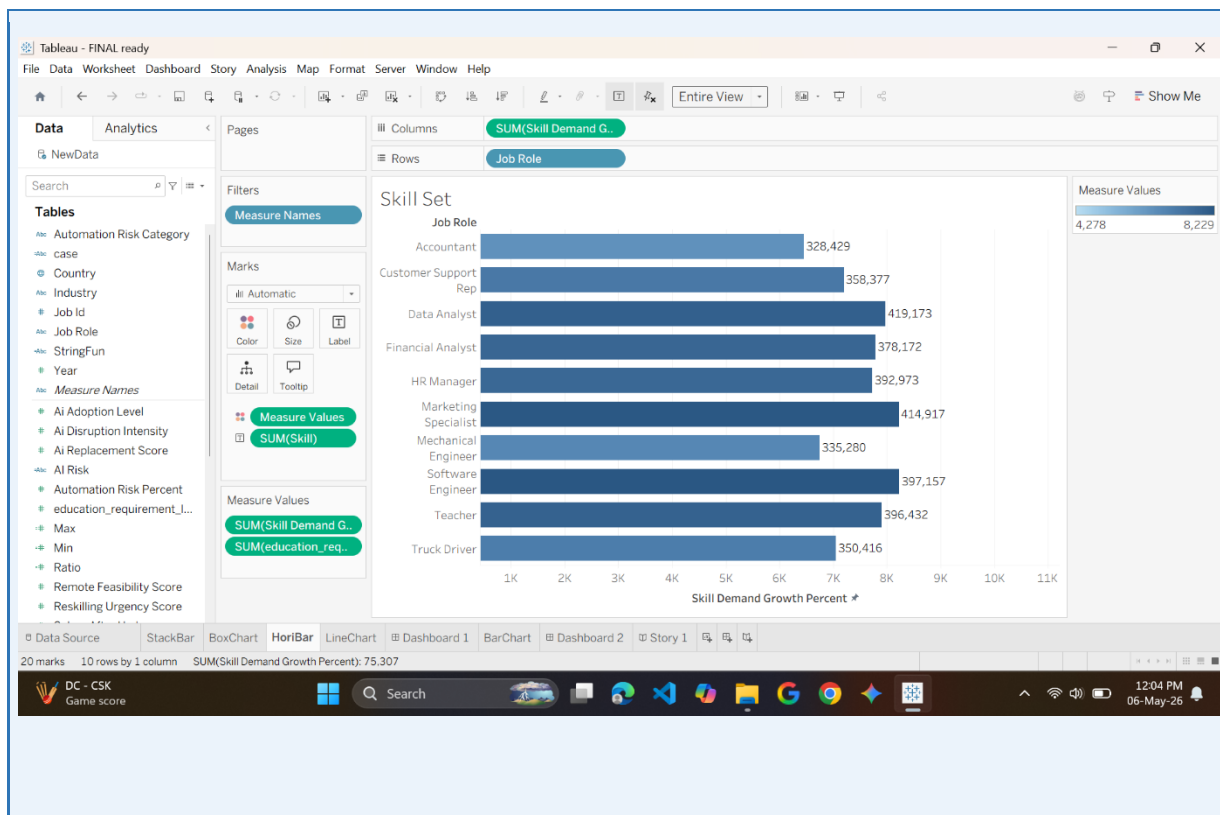
Definition: A Horizontal Bar Chart is a data visualization where categories are listed on the vertical axis and their corresponding values are represented by the length of horizontal rectangular bars. This format is particularly useful when category names are long (like "Marketing Specialist"), as it provides more space for text.

Purpose in Project: The purpose of this chart is to rank and compare the demand growth of different skills across various job roles. It allows for a quick visual comparison to see which roles are experiencing the highest growth (e.g., Data Analyst at 419,173 and Marketing Specialist at 414,917). The color gradient (darker blue for higher values) provides an additional visual cue to highlight the top-performing categories.

## Specifications:

X-Axis: Skill Demand Growth Percent (Quantitative value ranging from 0 to 11K on the scale, with specific data labels showing larger raw values).

Y-Axis: Job Role (Categorical labels including Accountant, Customer Support, Data Analyst, etc.).



## Grain 4 – Industry by Year Trend

Grain Description: The grain of this data is the annual measurement of reskilling urgency within specific sectors. It captures a time-series record for each industry, allowing for the tracking of how the "Reskilling Urgency Score" fluctuates from one year to the next.

## Chart 4 – Line Chart

Definition: A Line Chart is a type of chart which displays information as a series of data points called 'markers' connected by straight line segments. It is the primary tool used to visualize trends, patterns, and changes in data over a continuous interval of time.

Purpose in Project: The purpose of this chart is to track the progression of reskilling needs over time across multiple industries. It allows for Trend Analysis: Identifying if the need for reskilling is increasing, decreasing, or remaining stable (e.g., Finance shows a high initial score that fluctuates, while Manufacturing shows a sharp climb in the early 2020s).It enables

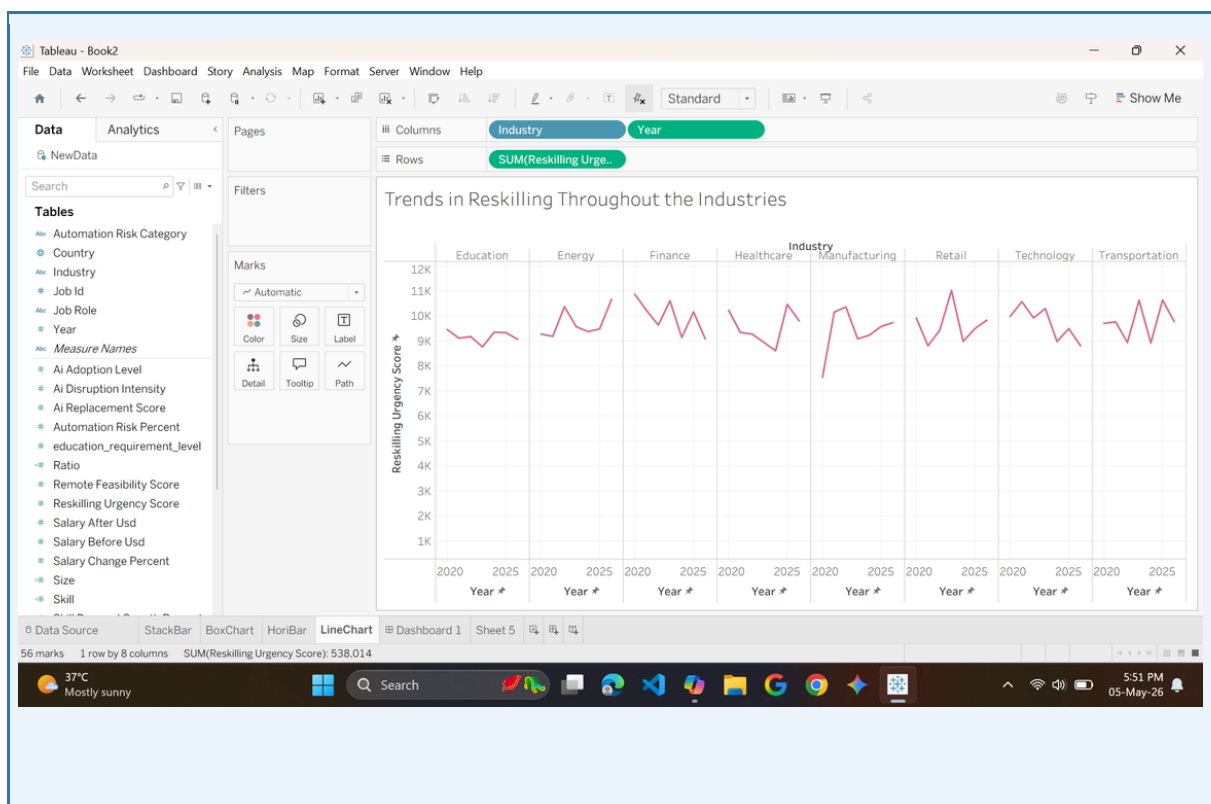
Cross-Industry Comparison: Using "small multiples" (facets) to see which industries are reacting more volatily to market changes compared to others.

Specifications:

X-Axis: Year (Temporal data ranging from approximately 2020 to 2025).

Y-Axis: SUM(Reskilling Urgency Score) (A quantitative measure ranging from 1K to 12K).

Columns (Facets): Industry (Education, Energy, Finance, Healthcare, Manufacturing, Retail, Technology, Transportation).



## Grain 5 – Job Role Risk by AI in Countries

Grain Description: The grain of this data is a specific professional role nested within a specific country. It captures the calculated AI impact score for each individual occupation (like "Truck Driver" or "Financial Analyst") for every country included in the dataset (Australia, Brazil, Canada, etc.).

## Chart 5 – Side-by-Side Bar Chart

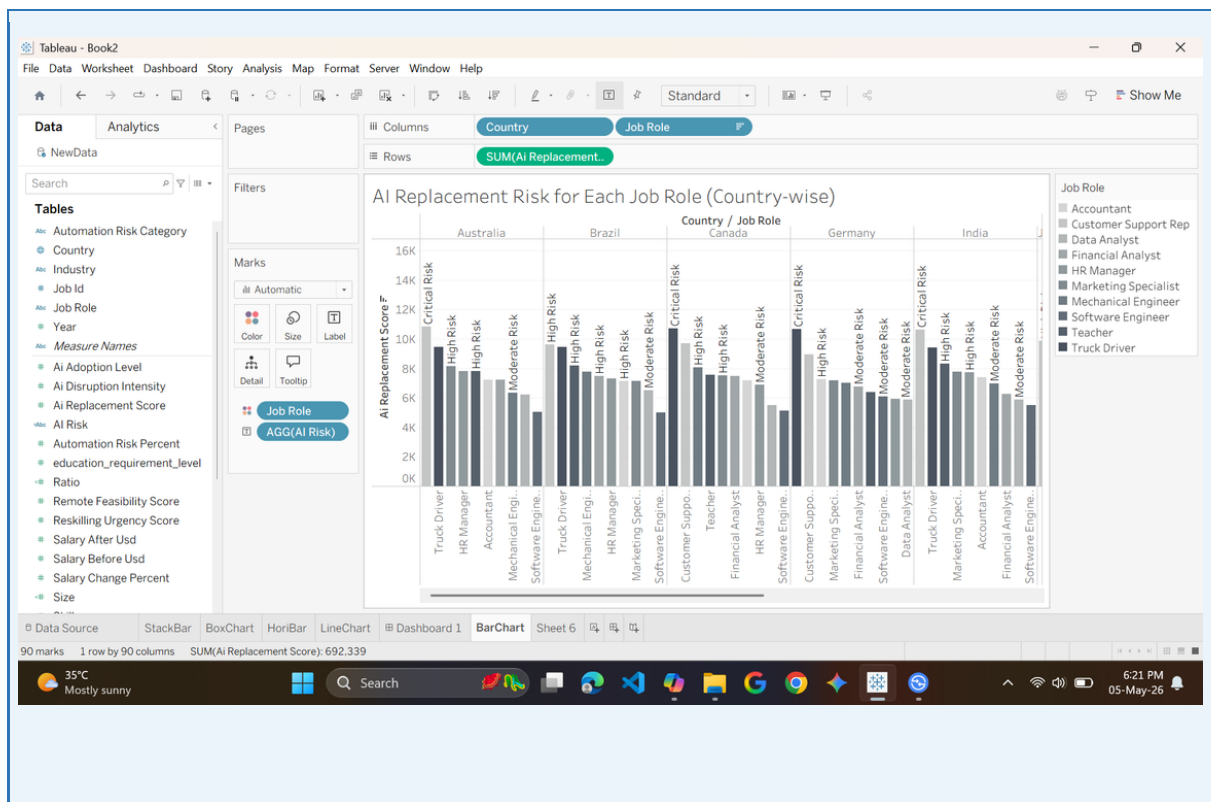
Definition: A Side-by-Side Bar Chart displays multiple bars for each primary category (in this case, Country) to compare different sub-groups (Job Roles) across them. Each bar's height represents a quantitative value, allowing for immediate comparison between both the groups and the sub-groups.

**Purpose in Project:** The purpose of this chart is to measure and compare the AI Replacement Score across different jobs and regions. It helps identify which specific roles are most at risk of AI replacement globally (e.g., Truck Driver consistently shows a very high score across multiple countries). It allows for Geographic Benchmarking, showing if a "Software Engineer" in Australia faces the same AI replacement risk as one in Germany.

**Specifications:**

**X-Axis:** Job Role (Nested under Country facets: Australia, Brazil, Canada, Germany, India).

**Y-Axis:** SUM(AI Replacement Score) (A quantitative measure ranging from 0K to 12K).

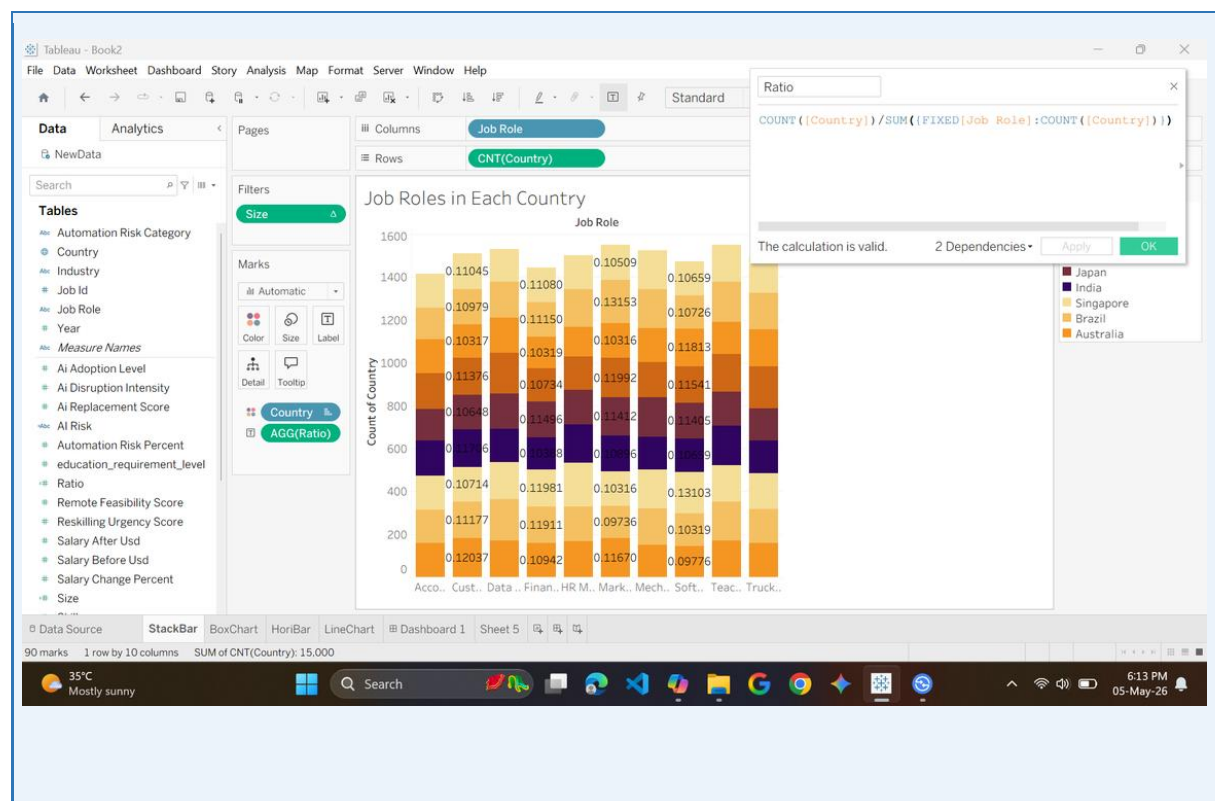


## 3.5 Calculated Fields

Definition: Calculated Fields in Tableau allow users to create new data fields using formulas applied to existing fields. They are written using Tableau's calculation language and can include functions for numbers, strings, logic, aggregation, and table-level computations. Calculated fields appear as new columns in the data panel and can be used just like any other field in the dataset.

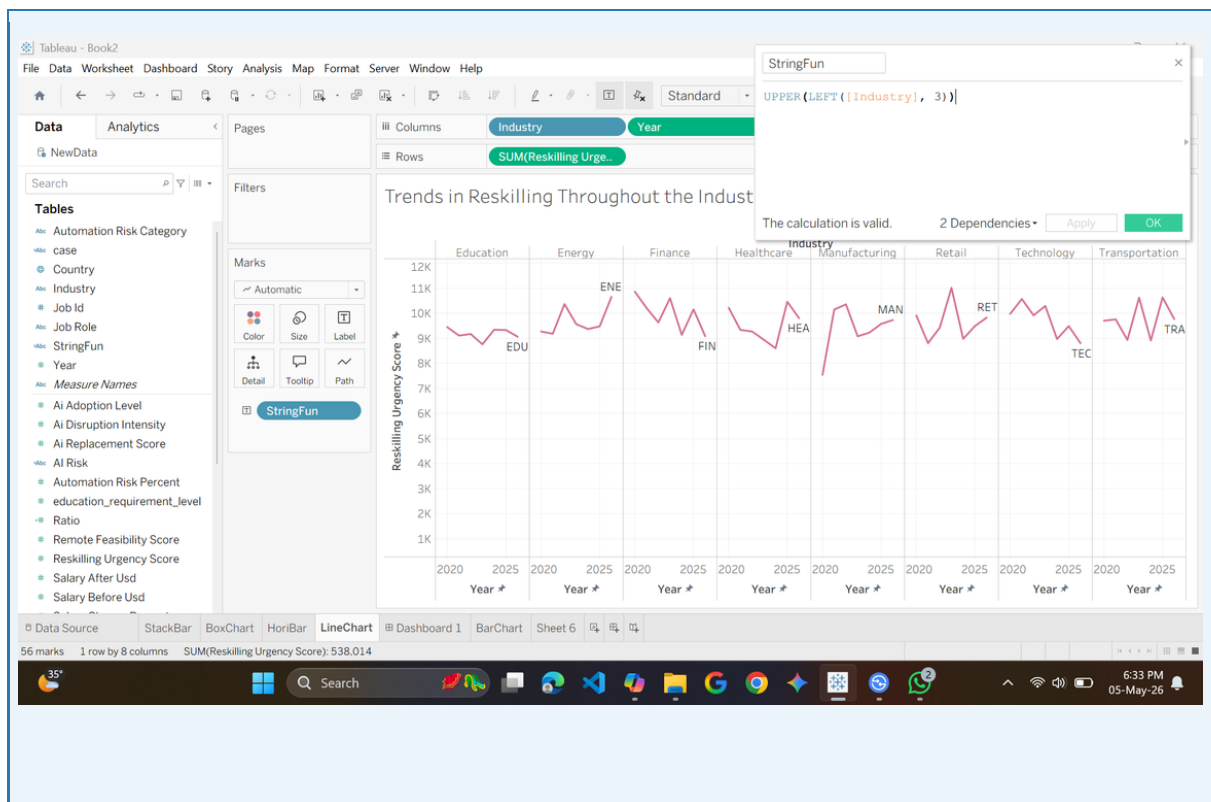
### 3.5.1 Number Function

Definition: Number functions in Tableau perform mathematical operations on numeric data fields. Common number functions include ROUND() (rounds to n decimal places), ABS() (absolute value), CEILING() (rounds up to nearest integer), FLOOR() (rounds down), POWER() (raises to a power), and SQRT() (square root).



### 3.5.2 String Function

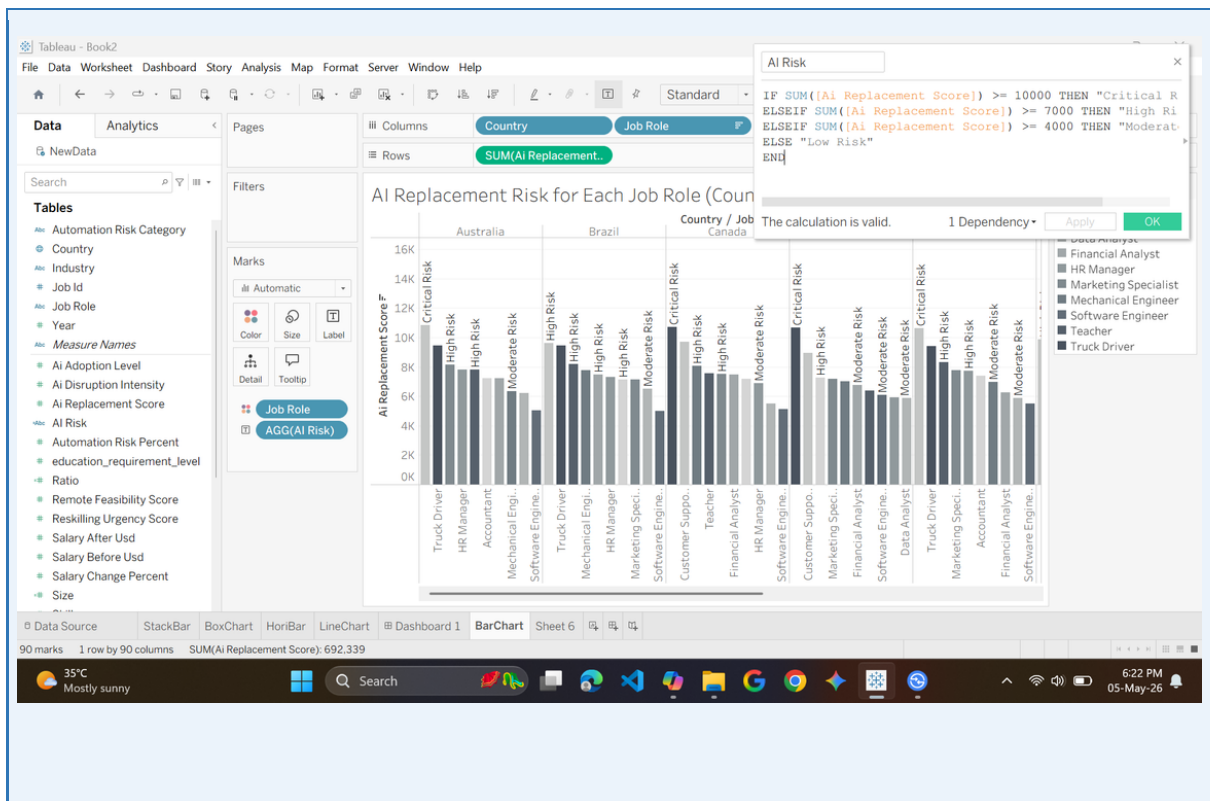
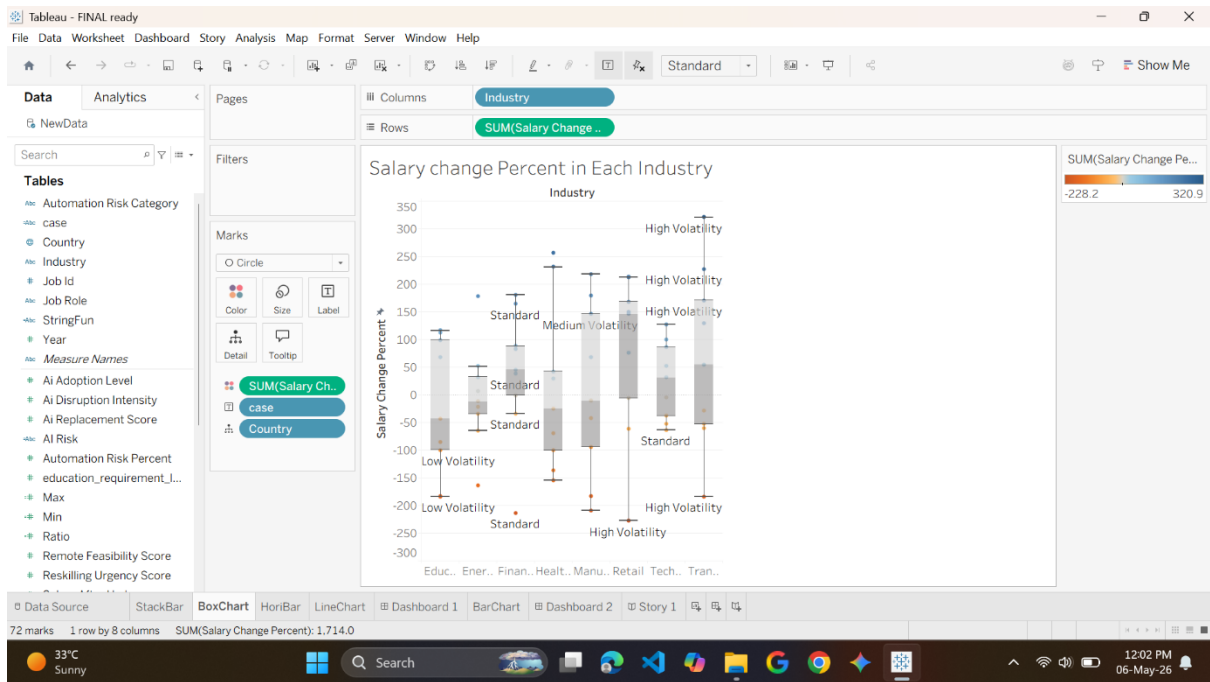
Definition: String functions in Tableau manipulate text data fields. Common string functions include UPPER() (converts to uppercase), LOWER() (converts to lowercase), LEN() (returns character count), TRIM() (removes leading/trailing spaces), LEFT() / RIGHT() / MID() (extracts substrings), and CONTAINS() (checks if a string includes a substring).



### 3.5.3 Logical Functions

Definition: Logical functions evaluate conditions and return results based on whether the condition is TRUE or FALSE. Tableau supports several logical operators and conditional expressions:

- CASE – Evaluates an expression against multiple values and returns a result for the first match. Like a switch statement.
- ELSEIF – Used in IF statements to evaluate multiple conditions in sequence without nesting multiple IF statements.
- IIF – Inline IF function. Takes three arguments: condition, true-result, false-result.  
Example: `IIF([Attrition]="Yes", 1, 0)`
- MAX / MIN – When used logically, return the maximum or minimum of two values. Can also compare dates.

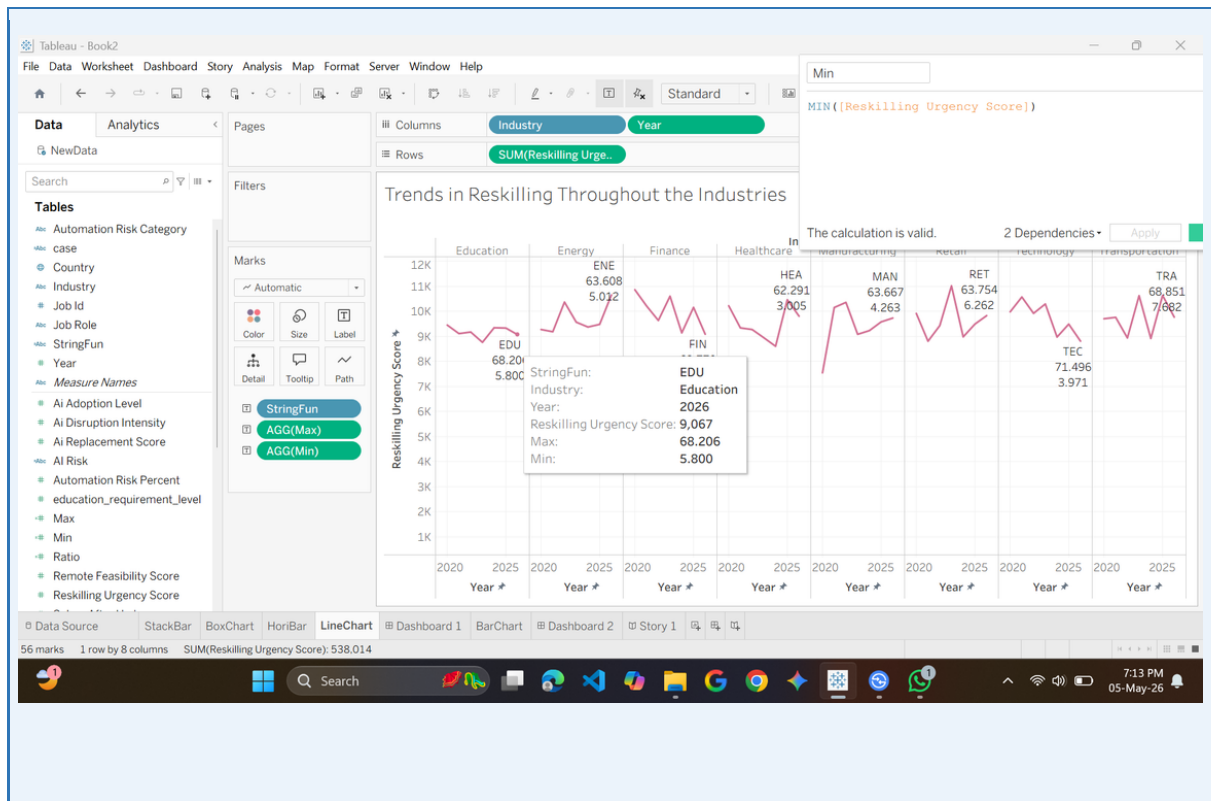


### 3.5.4 Aggregate Function

Definition: Aggregate functions in Tableau compute a single summary value from a group of rows. They are used to roll up data from row-level detail to a higher level of aggregation.

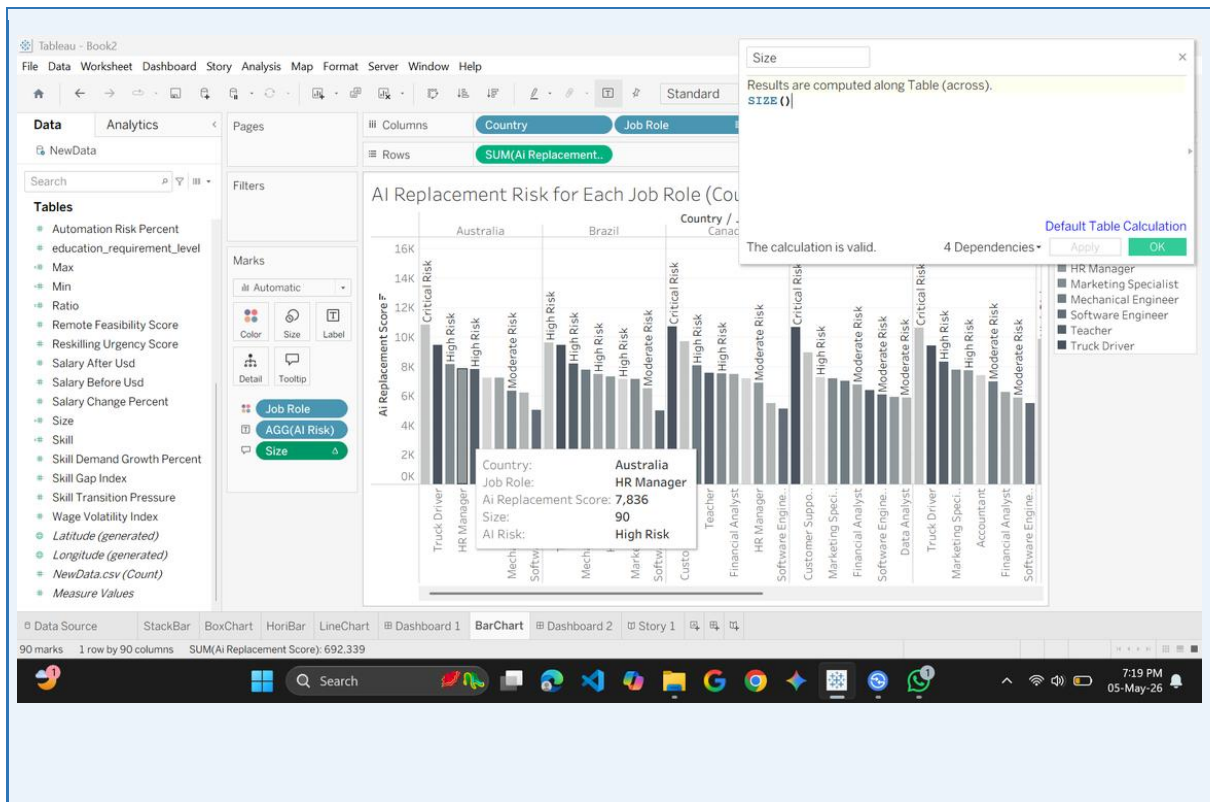
Common aggregate functions include SUM() (total), AVG() (average), COUNT() (number of

rows), COUNTD() (distinct count), MIN() (minimum value), and MAX() (maximum value).



### 3.5.5 Table Calculation

Definition: Table Calculations in Tableau are computed based on the values already in the visualisation (not the underlying database). They operate on the aggregated data displayed in the view. Common table calculations include Running Total (cumulative sum), Percent of Total (each value as a percentage of the overall total), Rank (ordering rows by value), and Difference (change from one row to the next).



## 3.6 KPI – Key Performance Indicators

Definition: A Key Performance Indicator (KPI) is a measurable value that demonstrates how effectively an organisation is achieving key business objectives. KPIs can be built using Measures (quantitative, numerical values) or Dimensions (categorical, qualitative attributes).

### KPI Using Measure

Definition – Measure: A Measure in Tableau is a quantitative field that can be aggregated (summed, averaged, counted, etc.). Measures are typically continuous numeric values used for calculations and comparisons.

KPIs Selected (Measures):

- Automation Risk Index: A numeric score representing the potential for a job to be automated.
- Wage Volatility: Numerical data indicating the fluctuation in earnings.
- Skill Reskilling Urgency: A calculated score or percentage reflecting the need for training.
- Equity Automation: Quantitative values likely related to automation impact levels.

### KPI Using Dimension

Definition – Dimension: A Dimension in Tableau is a qualitative field used to categorise, segment, or slice data. Dimensions are typically discrete text or date values such as Department, Gender, or Job Role. KPIs using dimensions show breakdowns or counts within specific categories.

KPIs Selected (Dimensions):

- Job Title: The name of the role (e.g., Data Analyst, Accountant).
  - Industry: The sector the job belongs to (e.g., Technology, Finance).
  - Country: The geographic location.
  - Year: While numeric, it is treated as a dimension to group or filter metrics by specific time periods.
  - Risk Level/Category: (e.g., Low, Medium, High) Categorical labels describing the intensity of a factor.
-

KPI :

### AI Replacement Risk

Customer support is most affected job due to AI

Software Engineer is the least affected due to AI

Energy sector requires ReSKILLING

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## 3.7 Dashboards

Definition: A Dashboard in Tableau is a consolidated view that combines multiple worksheets (charts), filters, and KPI cards into a single interactive canvas. Dashboards allow users to see different aspects of the data simultaneously and interact with the data through filters, parameters, and actions.

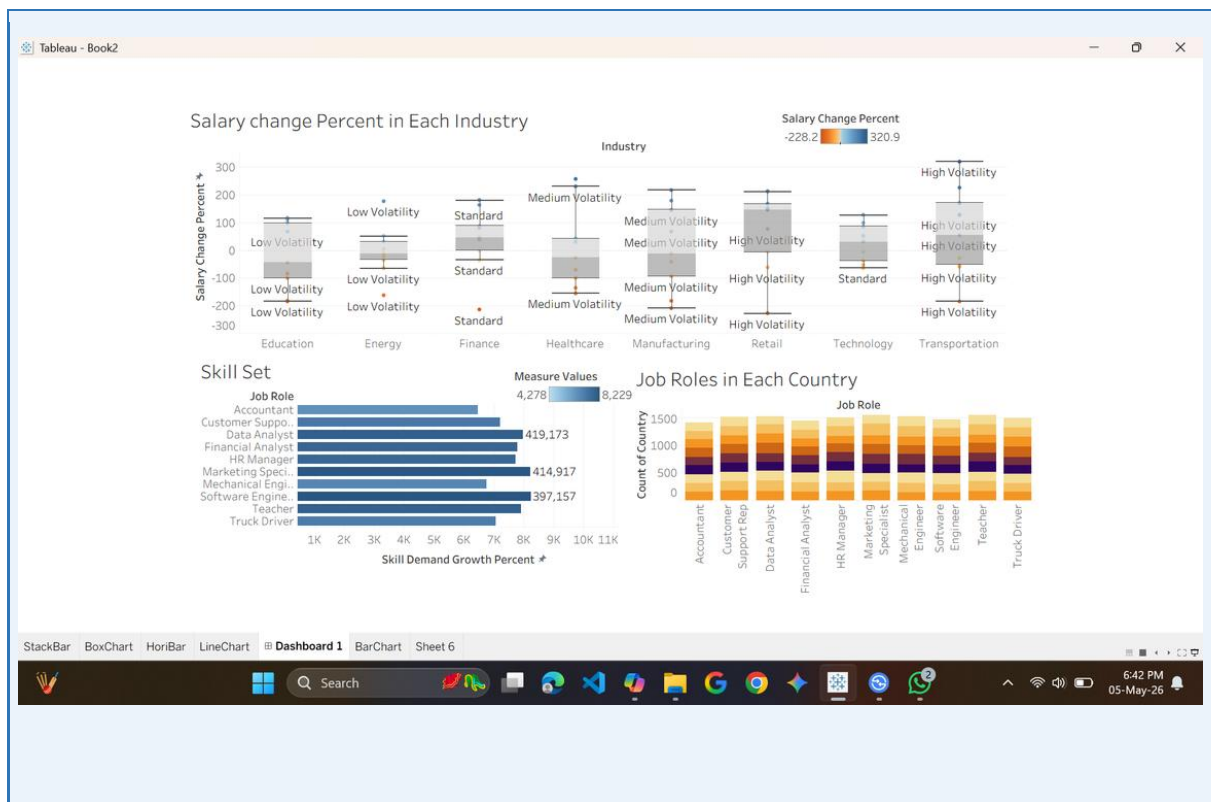
Types of Dashboards:

- Static Dashboard – The layout and displayed data are fixed and do not change based on user interaction. Filters are pre-applied and the view remains constant.
- Dynamic Dashboard – Contains interactive filters, parameters, or action filters that allow users to change what data is displayed in real time by clicking on elements or selecting options.

### Dashboard 1 – Salary change percent in each industry

Charts Included: Stack chart | Box chart | Horizontal Chart

Purpose: To analyse salary change percent in each industry

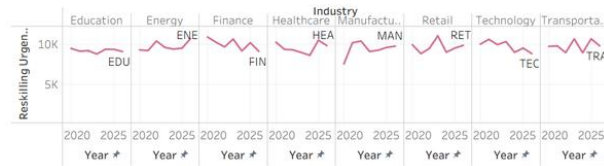


## Dashboard 2 – Trends in reskilling throughout the industries

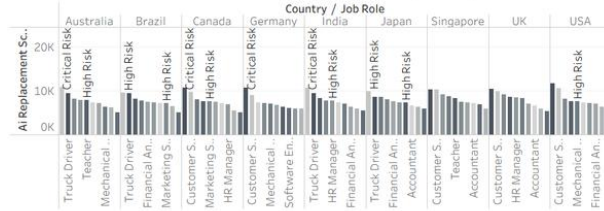
Charts Included: Line graph | Bar graph

Purpose: To analyse the trends in reskilling throughout the industries

## Trends in Reskilling Throughout the Industries



## AI Replacement Risk for Each Job Role (Country-wise)

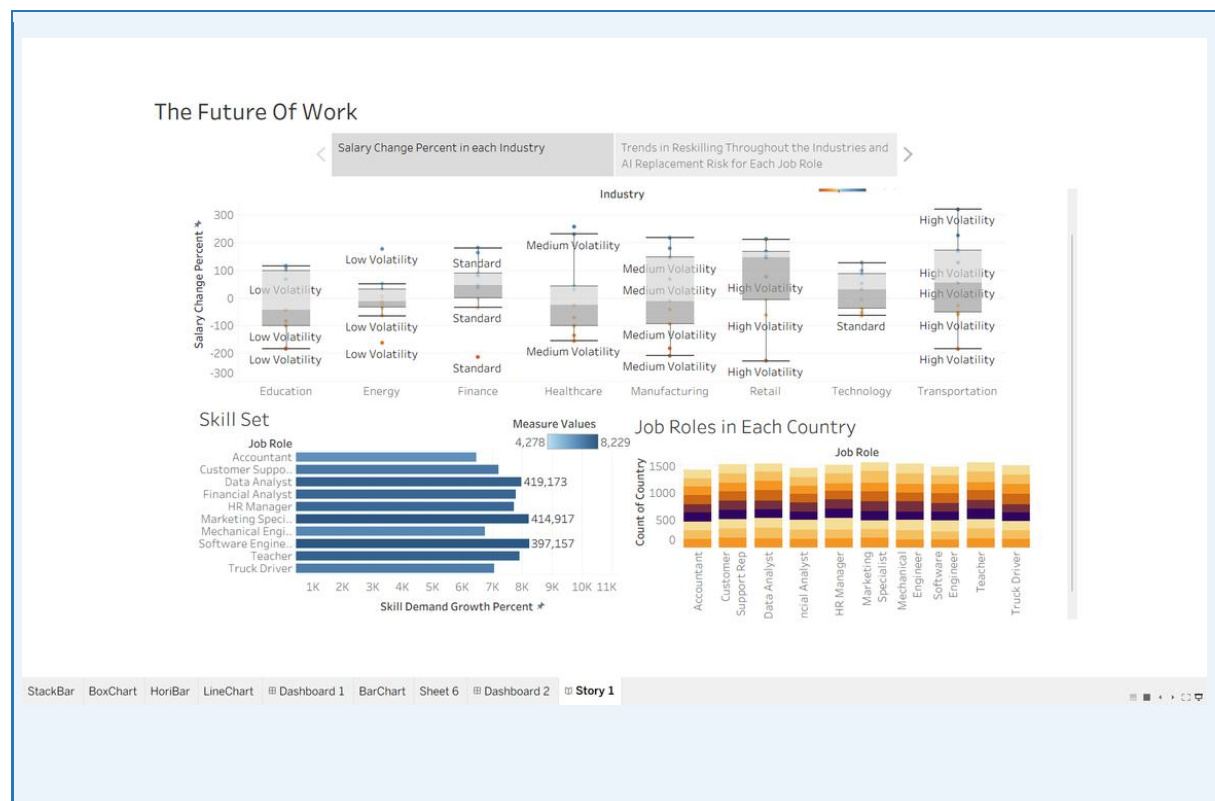


## 3.8 Story Board

Definition: A Story in Tableau is a sequence of visualisations or dashboards arranged as "story points" (slides) that together tell a narrative about the data. Each story point can contain a full dashboard or a single view with an optional caption. Stories are used to walk an audience through a logical progression of analysis, from initial observations to final conclusions and recommendations.

Narrative Caption: The Future Of Work

Key Message: The trend of ai influence over jobs



The Future Of Work

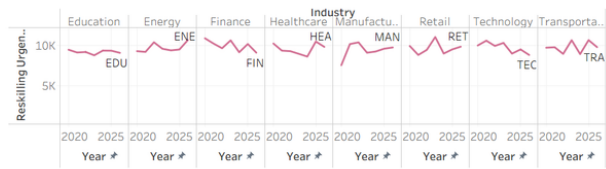
<

Salary Change Percent in each Industry

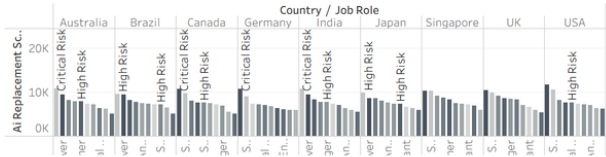
Trends in Reskilling Throughout the Industries and AI Replacement Risk for Each Job Role

>

Trends in Reskilling Throughout the Industries



AI Replacement Risk for Each Job Role (Country-wise)



## 4. Conclusion

From above analysis ,it is presented across various global markets confirms a definitive shift toward AI-integrated workflows. The correlation between high reskilling urgency and significant salary volatility suggests that the service market is currently in a state of 'productive disruption.' Ultimately, the findings suggest that 'AI literacy' has become the primary determinant of economic resilience in 2026.

— *End of Report* —